

Chain: randomized (2) $\rightarrow$ vanishing Infl at fixed $C^{(\alpha)}$

research/formal (working)

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Theorem 1 (Design $\rightarrow$ goal (iii) — schematic). *Fix classical level α and a family $C^{(\alpha)}$ satisfying joint fixed- θ validity (FV) on a valid support Θ_{val} (Part II Ass. FV; for score-threshold sets, $\Theta_{\text{val}} = \{\theta : F(\theta) \geq 1-\alpha\}$). Let A_r be the noisy-objective (or soft-argmin) map of `part_i_randomized_design.tex` with user budget $\eta_r \rightarrow 0$, Laplace scale $b_r = 2\varepsilon/\eta_r$ (or $\beta_r = \eta_r/(2\varepsilon)$), **restricted so that A_r and its oracle A_0 are supported on Θ_{val}** (nested: net of $[t_\alpha, 1]$; tradeoff: conservative $\Theta_0^\varepsilon \subseteq \{g^* \leq 0\}$), and concentration $(\varepsilon(n, \nu_r), \nu_r)$ from `instance_nested_uq.tex` or `instance_tradeoff_uq.tex` (DKW $\varepsilon = \sqrt{\log(2/\nu)/(2n)}$), via Ass. concentration. Then A_r is $(\eta_r, 0, \nu_r)$ -stable and (OV) holds at $q = \alpha$. If the fixed- θ map of $C^{(\alpha)}$ is unchanged by the noise in A_r , Part II Cor. vanish yields*

$$\mathbb{P}\{Y \notin C^{(\alpha)}(X; D, A_r(D))\} \leq e^{\eta_r} \alpha + \nu_r, \quad \text{Infl}_r \leq (e^{\eta_r} - 1)\alpha + \nu_r \rightarrow 0$$

along any sequence with $\eta_r \rightarrow 0$ and $\nu_r \rightarrow 0$ (e.g. $n \rightarrow \infty$ tightening concentration, or ν_r absorbed into a fixed Part II error budget).

Remark 1. Operational W5 answer to (iii): pay selection softness (large b_r) to buy small η_r ; keep classical level of $\mathcal{C}$ fixed; keep selection inside Θ_{val} so (OV) is not lost as $\eta \rightarrow 0$. Soft-on-all- $[0, 1]$ is stable but **not** a (iii) path for score-threshold $\mathcal{C}_\theta$.